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Lesson starter

Quadratic Equations

Factorable Quadratics

Name _____

S1

$x^2 - 6x = 0$

S2

$3x^2 - 9x = 0$

S3

$x^2 + 11x + 30 = 0$

S4

$3x^2 + 21x + 36 = 0$

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$4x^2 - 24x = 0$

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$x^2 - 3x - 10 = 0$

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Teacher answers & working

Quadratic Equations · Factorable Quadratics

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S1

$x^2 - 6x = 0$

Answer:

$x = 0, x = 6$

Working:

Step 1: Factor out the common x, then use the zero product property.

Step 2: Starting equation: $x^2 - 6x = 0$

Step 3: Factor out x:

$x(x - 6) = 0$

Step 4: Apply the zero product property: x = 0 or x -6 = 0

Step 5: Solve the second equation: x = 6

Step 6: Solutions: x = 0 and x = 6

Step 7: Final answer: x = 0, x = 6

S2

$3x^2 - 9x = 0$

Answer:

$x = 0, x = 3$

Working:

Step 1: Factor out x first, then solve the remaining linear factor.

Step 2: Starting equation: $3x^2 - 9x = 0$

Step 3: Factor out x:

$x(3x - 9) = 0$

Step 4: Apply the zero product property: x = 0 or 3x -9 = 0

Step 5: Solve 3x -9 = 0:

$3x = 9$

$x = 9/3 = 3$

Step 6: Solutions: x = 0 and x = 3

Step 7: Final answer: x = 0, x = 3

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S3

$x^2 + 11x + 30 = 0$

Answer:

$x = -6, x = -5$

Working:

Step 1: Find two numbers that multiply to c and add to b, then factorise.

Step 2: Starting equation: $x^2 + 11x + 30 = 0$

Step 3: We want two numbers that multiply to 30 and add to 11.

Step 4: Those numbers are 6 and 5.

So the factorisation is (x + 6)(x + 5) = 0.

$(x + 6) = 0$ or $(x + 5) = 0$

Step 5: Solutions: x = -6 and x = -5

Step 6: Final answer: x = -6, x = -5

S4

$3x^2 + 21x + 36 = 0$

Answer:

$x = -3, x = -4$

Working:

Step 1: Remove the common numerical factor first, then factorise the simpler quadratic.

Step 2: Starting equation: $3x^2 + 21x + 36 = 0$

Step 3: Factor out the greatest common factor 3:

$3(x^2 + 7x + 12) = 0$

Step 4: Since 3 is not 0, divide both sides by 3:

$x^2 + 7x + 12 = 0$

Step 5: Now factor the quadratic:

$(x + 3)(x + 4) = 0$

Step 6: Solutions: x = -3 and x = -4

Step 7: Final answer: x = -3, x = -4

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E1

$x^2 - 2x = 0$

Answer:

$x = 0, x = 2$

Working:

Step 1: Factor out the common x, then use the zero product property.

Step 2: Starting equation: $x^2 - 2x = 0$

Step 3: Factor out x:

$x(x - 2) = 0$

Step 4: Apply the zero product property: x = 0 or x -2 = 0

Step 5: Solve the second equation: x = 2

Step 6: Solutions: x = 0 and x = 2

Step 7: Final answer: x = 0, x = 2

E2

$4x^2 - 24x = 0$

Answer:

$x = 0, x = 6$

Working:

Step 1: Factor out x first, then solve the remaining linear factor.

Step 2: Starting equation: $4x^2 - 24x = 0$

Step 3: Factor out x:

$x(4x - 24) = 0$

Step 4: Apply the zero product property: x = 0 or 4x -24 = 0

Step 5: Solve 4x -24 = 0:

$4x = 24$

$x = 24/4 = 6$

Step 6: Solutions: x = 0 and x = 6

Step 7: Final answer: x = 0, x = 6

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E3

$x^2 - 3x - 10 = 0$

Answer:

$x = -2, x = 5$

Working:

Step 1: Find two numbers that multiply to c and add to b, then factorise.

Step 2: Starting equation: $x^2 - 3x - 10 = 0$

Step 3: We want two numbers that multiply to -10 and add to -3.

Step 4: Those numbers are 2 and -5.

So the factorisation is (x + 2)(x - 5) = 0.

$(x + 2) = 0$ or $(x - 5) = 0$

Step 5: Solutions: x = -2 and x = 5

Step 6: Final answer: x = -2, x = 5